

Initial Event Log Analysis – Guide to the Marimo notebook

The Marimo notebook integrates a set of patterns for initial event log analysis that are useful for exploring the raw event log data before doing any specific process mining analysis such as behavioral, conformance, performance, or deviance analysis.

After loading an event log, tab “Log Initialization” allows selecting which are the mandatory attributes. If present, by default “case:concept:name”, “time:timestamp”, and “concept:name” are selected as the case identifier, completion time, and activity name.

Moreover, in tab “Log initialization”, initial parameters can be set to fix the timestamp granularity and timezone, as well as to enforce a global ordering of events. The current ones are visible in the “Timestamp attribute detection” and “Granularity Analysis” tabs. These features implement patterns “Timestamp Granularity” and “Event Ordering”.

Since events logs may have duplicate events, these are detected and removed upon initialization, and a report of any removed duplicates is presented in the “Initialization Summary” tab, thereby realizing pattern “Duplicate Events Detection and Removal”.

Since large event logs may result in long computation times, tab “Time Frame” allows restricting the time frame from which to consider cases for analysis. Only cases fully contained within the time frame will be retained.

Log Initialization

Initialization Summary

Time Frame

Timestamp attribute detection

Granularity Analysis

Select a time zone to apply to all timestamp columns in the event log and case log:

Select Time Zone

UTC

(Optional) Select a granularity level to apply to all timestamp columns in the event log and case log:

Select granularity level

--

☐ Enforce global ordering of the event log by completion time

Select mandatory attributes from attributes that are fully filled:

CASE ID column

case:concept:name

COMPLETION TIME column

time:timestamp

ACTIVITY column

concept:name

Example: granularity analysis for attribute “time:timestamp” in the Sepsis event log:

Log Initialization

Initialization Summary

Time Frame

Timestamp attribute detection

Granularity Analysis

Showing the precision of 'time:timestamp' and whether any timestamp elements are constant, based on the data recorded therein:

Select a Timestamp attribute to inspect:

time:timestamp

Component analysis

Constant prefixes

Q Search...

Columns

Visualize

Explore

Export

Level str	Constant bool	Unique values int64	Value object	
year	False	3	None	
month	False	12	None	
day	False	31	None	
hour	False	24	None	
minute	False	60	None	
second	False	60	None	
millisecond	True	1	0	
sub-millisecond	True	1	0	
timezone	True	1	UTC	

9 rows, 4 columns

No selection

The following widget integrates features for enriching the event log, as well as for folding and unfolding activities, thereby implementing patterns “Enrichment”, “Activity Folding”, and “Activity Unfolding”.

Enricher

Manual Event Enrichment

Manual Case Enrichment

Activity Folding

Activity Unfolding

Log enrichment

Event log

Case log

1,050 cases · 15,213 events · 16 event types

applied

staged to add

staged to remove

EVENT ENRICHMENTS

LOCAL

Binning

not configured

Arithmetic

not configured

CONTEXTUAL

Time delta

not configured

Running aggregation

not configured

Leucocytes

3,382 events · 2 numeric attributes

EVENT TYPE

CUT

Leucocytes

a number

a timestamp

NUMERIC COLUMN

METHOD

Age

equal width

quantile

custom edges

EDGES

observed range 40 - 90

1 edge → 2 bins

18

x

+ Add

sorted automatically

INTERVAL

LABEL

EVENTS

(-∞, 18]

pediatric

0

(18, +∞)

adult

18

categorical result

The column holds a label, so on the case log it can only be tracked as latest, count, append or is_unique. The outer edges are unbounded, so a missing input stays missing rather than falling out of the last bin.

COLUMN NAME

age_group

Stage column

Age cut at [18] → Leucocytes.age_group

category · written on 3,382 events

EVENT LOG SCHEMA

33 columns

in sync

Leucocytes

4 attrs

org:group

cat

Age

num

Leucocytes

num

Diagnose

cat

CRP

4 attrs

LacticAcid

2 attrs

Admission NC

1 attrs

ER Triage

3 attrs

ER Registration

25 attrs

ER Cancic Triage

2 attrs

no staged changes

Case columns are not recomputed on apply.

Discard staged

Preview sample events

Apply enrichments

Enricher

Manual Event Enrichment

Manual Case Enrichment

Activity Folding

Activity Unfolding

Log enrichment

Event log

Case log

tab locked

1,050 cases · 15,213 events · 16 event types

applied

staged to add

staged to remove

EVENT ENRICHMENTS

LOCAL

Binning

✓ 1

Arithmetic

not configured

CONTEXTUAL

Time delta

not configured

Running aggregation

+1

Running aggregation

A value aggregated along each case as it goes.

all event types

written at every event position

NUMERIC COLUMN

Leucocytes

AGGREGATE

cumulative sum

count

running max

running min

forward fill

Every event carries the last Leucocytes value seen in the case so far. Events before the first one stay NaN —there is no last value yet, and 0 would be a reading nobody took.

COLUMN NAME

leucocytes_last_seen_2

Stage column

ffill(Leucocytes) per case → every event.leucocytes_last_seen_2

float64 · written on every event

Missing before the first value

Events before the case's first Leucocytes value get NaN rather than 0 —the last value seen has no answer yet, and 0 would invent one.

RUNNING AGGREGATION COLUMNS ON THIS LOG

event.leucocytes_last_seen

last value seen of Leucocytes per case

x

EVENT LOG SCHEMA

35 columns

+1 staged

Leucocytes

6 attrs

✓ 1

+1

org:group

cat

Age

num

Leucocytes

num

Diagnose

cat

age_group

x

leucocytes_last_seen

x

CRP

5 attrs

+1

LacticAcid

3 attrs

+1

Admission NC

2 attrs

+1

ER Triage

4 attrs

+1

+1 columns staged

Case columns are not recomputed on apply.

Discard staged

Preview sample events

Apply enrichments

Tab “Enricher” provides an interactive widget to enrich the event log without having to write any code. It is possible to enrich both the event log and the corresponding case log derived from it. Note that the case log derivation is automatic upon log loading and independent of whether case enrichments are applied; this derivation realizes pattern “Case Table”.

In tab “Enricher”, sub-tab “Event log”, “Binning” allows deriving an attribute based on the values of a selected existing attribute. In the first example shown above, the value of attribute “Age” from activity “Leucocytes” is used to populate a new attribute “age_group” with values “pediatric” and “adult”, using value 18 as the splitting value. In the second example shown above, a running aggregation is applied to forward-fill the value of attribute “Leucocytes”.

With sub-tab “Case log” it is possible to apply case-level enrichments. In the example shown above, “Attribute tracking” allows deriving a case-level attribute that informs whether the value of attribute “Age” is unique within the case.

Custom enrichments at both event and case levels are possible by writing Python code in the respective tabs, as shown in the example above where weekday enrichment code is provided. The effect of this (generic) enrichment is the addition of a new attribute that makes the weekday encoded in attribute “time:timestamp” explicit.

Similar to the manual enrichment, tabs “Activity Folding” and “Activity Unfolding” allow writing custom Python code to fold, resp. unfold a given activity. In the example provided, activities “Release \$code” in the Sepsis log are folded into a common activity “Release”, where the \$code is added to the activity schema as a new attribute, preserving the information that was previously encoded in the activity name. Upon applying a folding, resp. an unfolding, the results of the application are shown to the user, as demonstrated in the screenshot below.

Fold activity types whose distinction is not needed into a common activity type, optionally keeping the distinction in a new attribute. The folded log replaces the event log for all analyses below.

Define fold_activities(df)

```

1 def fold_activities(df):
2     # Relevant for the Sepsis event log: fold "Release <letter>" into "Release",
3     # keeping the letter in a new attribute ReleaseCode.
4     release_code = df[ACTIVITY].str.extract(r'^Release ([A-Za-z])$')[0]
5
6     df['ReleaseCode'] = release_code
7     df.loc[release_code.notna(), ACTIVITY] = 'Release'
8
9     return df
10

```

Apply folding

Activity types after folding: 12 (before: 16). Added attributes: ['ReleaseCode'].

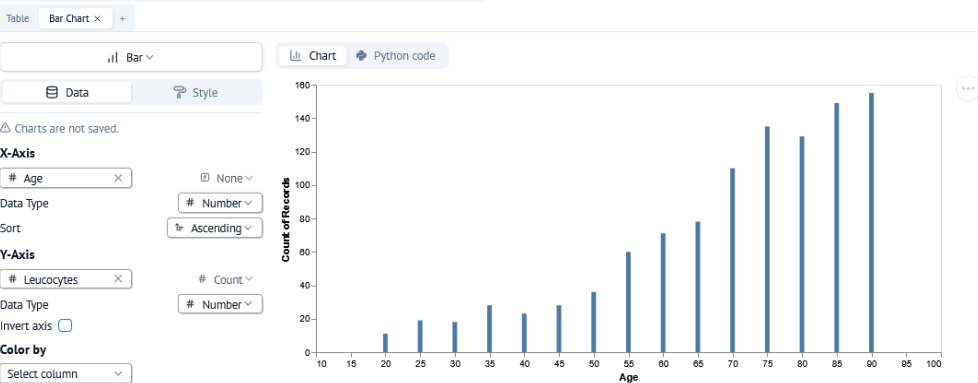
conceptname str	Events before int64	Events after int64	Change str unique: 3
Release	0	782	new
Release A	671	0	removed
Release B	56	0	removed
Release C	25	0	removed
Release D	24	0	removed
Release E	6	0	removed
Leucocytes	3,382	3,382	unchanged
CRP	3,262	3,262	unchanged
LacticAcid	1,466	1,466	unchanged
Admission NC	1,182	1,182	unchanged

In the “Events View” section, the widget allows inspecting both the event log and the case log, potentially focusing only on global or local attributes. This widget allows inspecting in detail the attributes, also graphically by selecting “Visualize” in the top-right corner and defining a custom chart, thereby realizing patterns “Univariate Analysis,” “Bivariate Analysis”.

Below are the view of the raw event log with all events in tabular form, and a bar chart:

Index0 int64	Hypotensie object (nulls: 14163) unique:	SIRS CritTachypnea object (nulls: 14163) unique:	Hypoxie object (nulls: 14163) unique:	Oligurie object (nulls: 14163) unique:	conceptname str unique: 12	DiagnosticUrinarySediment object (nulls: 14163) unique:	DiagnosticThorax object (nulls: 14163) unique:
0	True	True	False	False	ER Registration	True	True
1	None	None	None	None	Leucocytes	None	None
2	None	None	None	None	CRP	None	None
3	None	None	None	None	LacticAcid	None	None
4	None	None	None	None	ER Triage	None	None
5	None	None	None	None	ER Sepsis Triage	None	None
6	None	None	None	None	IV Liquid	None	None
7	None	None	None	None	IV Antibiotics	None	None
8	None	None	None	None	Admission NC	None	None
9	None	None	None	None	CRP	None	None

Global Attributes Local Attributes Case Inspection Case Log Raw Events



In section “Activity Schema View”, one can inspect the activity schemas, their overlaps, and statistics about activities and how attributes are classified based on their population. This widget therefore realizes patterns “Activity Schema Definition and Overlap”, “Schema Population”, “Activity-Attribute Association”, “Attribute Classification”, and “Activity Occurrence Statistics”.

In the example below, the activity schemas for the Sepsis event log are shown, for example making it explicit that activity “Leucocytes” has in addition to the mandatory attributes also “Age”, “Diagnose”; “Leucocytes”, and “age:group”:

Select an activity type to inspect the attributes of its schema:

Select activity Leucocytes

Activity Schemas			
Schema Overlaps			
Attributes			
Selected Activity			
Q Search... Columns Visualize Explore Export			
concept:name str	incidence int64	extra_schema_keys object unique:	missing_values object unique:
Leucocytes	3,382	["Age","Diagnose","Leucocytes","age_group"]	{"Age":-3364,"Leucocytes":-22,"Diagnose":-3366,"age..."}
CRP	3,262	["Age","CRP","Diagnose"]	{"CRP":-139,"Age":-3252,"Diagnose":-3257}
LacticAcid	1,466	["LacticAcid"]	{"LacticAcid":-12}
Admission NC	1,182		{}
ER Triage	1,053	["Age","Diagnose"]	{"Age":-1047,"Diagnose":-1049}
ER Registration	1,050	["Age","Diagnose","DiagnosticArtAstrup","DiagnosticBL..."]	{"Age":-55,"Diagnose":-296}
ER Sepsis Triage	1,049	["Age","Diagnose"]	{"Age":-1042,"Diagnose":-1044}
IV Antibiotics	823		{}
Release	782	["ReleaseCode"]	{}
IV Liquid	753	["Age","Diagnose"]	{"Age":-739,"Diagnose":-740}
12 rows, 3 columns			
10 / page << < Page 1 of 2 > >>			
No selection			

While attributes “Age”, “Diagnose” and “folded_data” are shared between multiple activity schemas. Note that “folded_data” was generated with the generation of the case table as an additional attribute of the event log.

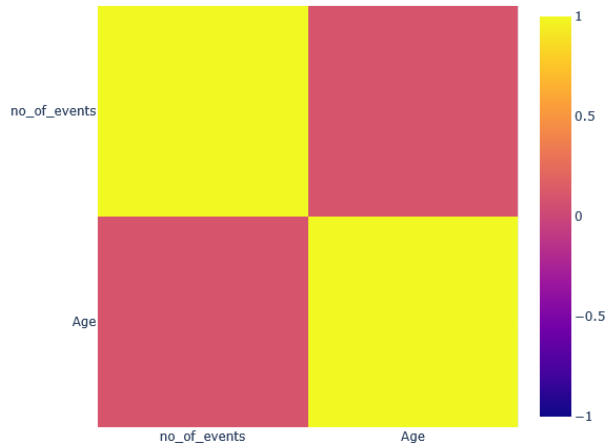
Activity Schemas			
Schema Overlaps			
Attributes			
Selected Activity			
Q Search... Columns Visualize Explore Export			
concept:name str	Age int64	Diagnose int64	folded_data int64
Leucocytes	18	16	3,360
CRP	10	5	3,124
LacticAcid	0	0	1,454
ER Triage	6	4	6
ER Registration	995	754	1,050
ER Sepsis Triage	7	5	7
IV Liquid	14	13	14
7 rows, 3 columns			
No selection			

The widget also shows how attributes are classified and populated in the log:

Activity Schemas			
Schema Overlaps			
Attributes			
Selected Activity			
Q Search... Columns Visualize Explore Export			
Attribute str unique: 36	Classification str unique: 4	Level str unique: 2	Events populated int64
Hypotensile	non-standard, exclusive	case	1,050
SIRSCritTachypnea	non-standard, exclusive	case	1,050
Hypoxie	non-standard, exclusive	case	1,050
Oligurie	non-standard, exclusive	case	1,050
concept:name	mandatory	event	15,213
DiagnosticUrinarySediment	non-standard, exclusive	case	1,050
DiagnosticCthorax	non-standard, exclusive	case	1,050
DiagnosticUrinaryCulture	non-standard, exclusive	case	1,050
DiagnosticArtAstrup	non-standard, exclusive	case	1,050
LacticAcid	non-standard, exclusive	event	1,454
36 rows, 10 columns			
10 / page << < Page 1 of 4 > >>			
No selection			

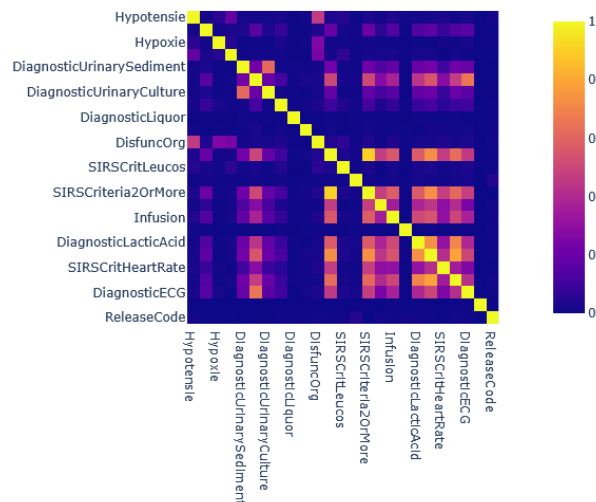
The notebook allows also analyzing attribute dependencies and functional relationships, thereby realizing pattern “Attribute Functional Relationship” and further supporting pattern “Bivariate Analysis”. On the one hand, this is done by showing the pairwise Pearson correlation between numeric attributes:

Pairwise Pearson correlation between the numeric case attributes:



On the other hand, the normalized mutual information is provided for all pairs of attributes, regardless of whether they are numerical or categorical:

Pairwise normalized mutual information between all case attributes (numeric attributes discretized into 10 quantile bins):



Functional relationships are displayed for all pairs of attributes, and selected relationships can be inspected in detail, thereby supporting the identification of redundant information and realizing pattern “Redundant attribute”:

Select attribute pair Hypotensie <-> timestamp (one-to-many)

All functional relationships Selected mapping Not functional

'Hypotensie' <-> 'timestamp': **one-to-many** (2 distinct 'Hypotensie' values, 1050 distinct 'timestamp' values, support 0.0690 over 1050 events)

Functional mapping 'timestamp' -> 'Hypotensie':

Table	
Q Search...	
timestamp datetime64[us, UTC] Nov 7, 2013-Feb 26, 2015	Hypotensie bool True 54 False 996
2014-10-22 09:15:41+00:00	True
2014-12-21 10:04:24+00:00	False
2014-02-09 18:29:29+00:00	False
2014-07-10 09:52:00+00:00	False
2015-01-15 19:14:58+00:00	False
2014-05-07 16:56:57+00:00	False
2014-09-05 12:40:47+00:00	False
2014-03-11 08:50:02+00:00	False
2013-11-09 08:21:03+00:00	False
2014-01-02 15:45:45+00:00	False
1,050 rows, 2 columns	10 / page << < Page 1 of 105 > >> No selection

The notebook also shows any detected batch event candidates and transaction candidates, based on the minimum thresholds that can be set by the user, thereby realizing patterns “Super Event and Batching Candidates” and “Transaction Candidates”:

Super event candidates

The widget below allows inspecting any candidates identified for super events in the log.

Minimum set size for super-event detection 2

Q Search...					Columns	Visualize	Explore	Export
event_set	case_support	set_coverage	event_coverage_avg	occurrences				
object	float64	float64	float64	int64				
unique:								
["CRP";"Leucocytes";"LacticAcid"]	0.805714285714287000	0.9837209302325581000	0.5788234949088251000	1,352				
["CRP";"Leucocytes"]	0.6866666666666666000	0.7166998011928429000	0.4974530650781846500	1,652				
["IV Antibiotics";"IV Liquid"]	0.045714285714287140	0.0637450199203187200	0.0610341138483732100	48				
["Leucocytes";"LacticAcid"]	0.013333333333333340	0.0162790697674418600	0.0068446788753234160	14				
["CRP";"LacticAcid"]	0.0047619047619047620	0.0058139534883720930	0.0029660658975193283	6				
["CRP";"Leucocytes";"IV Antibiotics";"LacticAcid"]	0.0019047619047619048	0.0025348542458808617	0.0012497192444881123	2				
["CRP";"Leucocytes";"Admission IC";"LacticAcid"]	0.0009523809523809524	0.0092592592592592590	0.0024578450518272987	1				
["CRP";"Leucocytes";"IV Liquid";"LacticAcid"]	0.0009523809523809524	0.0013755158184319120	0.0006530982271601552	1				
["CRP";"Leucocytes";"Admission NC"]	0.0009523809523809524	0.0012610340479192938	0.0004827557029515965	1				
["CRP";"Admission NC"]	0.0009523809523809524	0.0012594458438287153	0.0005762920405302924	1				
11 rows, 5 columns					10 / page << Page 1 of 2 >> No selection			

Batch event candidates

The widget below allows inspecting any candidates identified for batching in the log.

Minimum distinct cases for batch-event detection 2

Q Search...

Columns

Visualize

Explore

Export

conceptname

str

unique: 5

timestamp

datetime64[us, UTC]

Nov 14, 2013-Feb 24, 2015

case_count

int64

2 3 6 9

case_ids

object

unique:

CRP

2014-04-20 07:00:00+00:00

9

["AFA";"CR";"GT";"MEA";"NEA";"NF";"WW";"YLA";"ZH"]

Leucocytes

2014-08-25 06:00:00+00:00

9

["BX";"DGA";"KH";"NGA";"QMA";"SX";"WDA";"WGA";"YKA"]

CRP

2014-01-08 07:00:00+00:00

8

["AY";"HDA";"OHA";"QM";"TU";"VU";"WLA";"XI"]

CRP

2014-08-25 06:00:00+00:00

8

["BX";"DGA";"KH";"NGA";"QMA";"SX";"WDA";"WGA"]

CRP

2014-04-18 07:00:00+00:00

7

["GT";"NF";"PLA";"SI";"YK";"YLA";"ZH"]

CRP

2014-06-07 06:00:00+00:00

7

["CY";"DB";"GF";"KC";"ODA";"YO";"ZF"]

CRP

2014-08-05 06:00:00+00:00

7

["AI";"BS";"DZ";"RBA";"ST";"UD";"WP"]

CRP

2014-08-10 06:00:00+00:00

7

["BF";"BS";"NGA";"NN";"RZ";"TA";"UD"]

CRP

2014-08-12 06:00:00+00:00

7

["BGA";"GN";"HNA";"NN";"RZ";"TAA";"WT"]

CRP

2014-08-23 06:00:00+00:00

7

["DGA";"IS";"KK";"KS";"OKA";"QMA";"YKA"]

1,148 rows, 4 columns

10 / page

<< < Page 1 of 115 > >>

No selection

Transaction detection

The widget below allows inspecting any candidates identified for transactions in the log. Events can be excluded from the detection of transaction candidates by specifying their activity type.

Excluded events

Select... ▾

Similarity threshold

0.95 ▴ ▾

Minimum set size

2 ▴ ▾

Q Search...

Columns Visualize Explore Export

event_set object	set_size int64	similarity float64	
["ER Registration","ER Sepsis Triage","ER Triage"]	3	0.9974666485777597	

1 row, 3 columnsNo selection

Finally, users can formalize properties that they hypothesize hold in the log to verify whether their expectations are met. This is done by manually writing custom Python code that encodes the property in a dedicated widget. An example for the Sepsis event log is provided to check whether attribute “SIRSCriteria2OrMore” takes value true when at least two SIRS attributes take value true. If the property is violated, all cases that violate it are reported:

Define check_rule(df):

```
1 def check_rule(df):
2     # Relevant for the Sepsis event log: attribute SIRSCriteria2OrMore is expected to be True if at least
3     # two of the attributes SIRSCritHeartRate, SIRSCritLeucos, SIRSCritTachypnea, and SIRSCritTemperature are True.
4
5     criteria = ['SIRSCritHeartRate', 'SIRSCritLeucos', 'SIRSCritTachypnea', 'SIRSCritTemperature']
6
7     # events where the attributes are not populated cannot be evaluated and are not violations
8     evaluable = df[criteria + ['SIRSCriteria2OrMore']].notna().all(axis=1)
9     populated = df[evaluable]
10
11     expected = populated[criteria].astype(bool).sum(axis=1) >= 2
12     return populated[expected != populated['SIRSCriteria2OrMore'].astype(bool)]
13
```

Check rule

Q Search...

Columns Visualize Explore Export

Violations reported int64	Cases violating int64	Cases in log int64	Cases violating % float64	
29	29	1,050	2.76	

1 row, 4 columnsNo selection

Violating cases (29): ['H', 'HA', 'BB', 'RG', 'VH', 'HI', 'QI', 'QK', 'BO', 'HP', 'PP', 'UQ', 'PR', 'YW', 'AX', 'PX', 'GZ', 'PCA', 'REA', 'JFA', 'JGA', 'AJA', 'MJA', 'AKA', 'FKA', 'MLA', 'WLA', 'GMA', 'CNA']

Q Search...

Columns Visualize Explore Export

Index0 int64	Hypotensie object unique:	SIRSCritTachypnea object unique:	Hypoxie object unique:	Oligurie object unique:	concept:name str unique: 1	DiagnosticUrinarySediment object unique:	DiagnosticXthorax object unique:
91	False	True	False	False	ER Registration	False	False
373	False	True	False	False	ER Registration	False	False
580	False	True	False	False	ER Registration	False	False
2,660	False	True	False	False	ER Registration	False	False
3,048	False	False	False	False	ER Registration	False	False
3,184	False	True	False	False	ER Registration	True	True
3,284	False	True	False	False	ER Registration	False	False
4,203	False	True	False	False	ER Registration	False	True
5,684	False	False	False	False	ER Registration	False	False
6,163	False	False	False	False	ER Registration	False	False
6,770	False	True	False	False	ER Registration	False	False